

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory**Lab 1 – C Programming Revision****Date:** 31.07.2026**Semester:** B.Tech 3rd Semester**Section:** CSE B1**Instructions**

- Write separate programs for each question.
- Use user-defined functions wherever applicable.
- Follow proper coding standards, meaningful variable names, and clean indentation.

1. Array Statistics

Write a program to accept **N** integers in an array and display:

- Largest element
- Smallest element
- Average of all elements

2. Reverse an Array

Write a program to accept **N** integers in an array and display the elements in reverse order.

3. Linear Search

Write a program to accept **N** integers in an array and search for a given element. Display its position if found; otherwise, display **"Element not found"**.

4. Swap Using Pointers

Write a program to swap two integers using pointers and a user-defined function.

5. Sum of Array Using Pointer Arithmetic

Write a program to accept **N** integers and calculate the sum of all elements using pointer arithmetic instead of array indexing. ***Hint:** Traverse the array using a pointer instead of `arr[i]`.*

6. Student Structure

Create a structure **Student** containing:

- Roll Number
- Name
- Marks

Accept details of **5 students** and display all the records in a neat tabular format.

7. Student Analysis

Using the same `Student` structure from the previous question, write a program to:

- Display the student who secured the highest marks.
- Calculate and display the average marks of the class.

8. Dynamic Memory Allocation

Write a program to:

- Accept `N` from the user.
- Allocate memory dynamically using `malloc()`.
- Store `N` integers.
- Display all the elements.
- Free the allocated memory before exiting.

***Hint:** Use `malloc()` to allocate memory and `free()` to release it after use.*

9. Pointer to Structure

Create a structure `Student` containing Roll Number, Name, and Marks. Declare a pointer to the structure and use the `->` operator to:

- Accept student details.
- Display student details.

***Hint:** Use the `->` operator to access structure members through a pointer.*

10. Dynamic Student Records

Create a structure `Student` containing Roll Number, Name, and Marks. Dynamically allocate memory for `N` students using `malloc()`. Accept the details of all students and display the student who secured the highest marks. Free the allocated memory before exiting. ***Hint:** Allocate memory for an array of `Student` structures using `malloc()` and traverse the records to identify the student with the highest marks.*